

Splitting the Difference — Instructor Notes

Class 10 · expected value vs. distribution · paper and coins, no notebook

The idea. A procedure can be perfectly fair *in expectation* and still hand out lopsided outcomes almost every time. The inventory is rigged to make this unmissable: the lake cabin (\$60,000) is worth exactly as much as every other item combined. Whoever wins that one flip is ahead no matter how the other eleven fall, so the aunt's share is bimodal, with humps near \$30,000 and \$90,000 and a nearly empty middle right where the expected value sits.

What students produce. A prediction, two settlements per team, a class histogram, and written answers to eleven questions. **What this does not claim to teach:** how to compute a distribution formally, or fair-division theory. 4.4 touches that theory, but only as discussion.

Materials. One coin per team. A pre-drawn 12-bin histogram on the board (\$0–120k in \$10k bins, matching the handout). Optional: the projection code in the appendix.

Timing (50 min)

Segment	Min	What happens
Setup + Part 1	5	Read the scenario aloud. Students commit to 1.1 and 1.2 in ink. Tally the 1.2 answers on the board <i>before</i> any flipping.
Part 2	10	24 flips per team (12 items $\times$ 2 rounds), add up totals, post both aunt totals.
Part 3	12	Build the histogram, answer 3.1–3.5. Reveal the exact distribution (below).
Part 4	21	Discussion. 4.1 and 4.5 are the ones that matter most; 4.3 carries the quantitative point.
Buffer	2	

Watch out. The 50-minute budget is an estimate. Trim 4.2 or 4.4 first if you're short on time; keep 4.1, 4.3 and 4.5.

The numbers (exact: all $2^{12} = 4096$ equally likely settlements enumerated)

	Original	4.3 variant (cabin $\rightarrow$ $4 \times \$15k$)
Expected aunt share	\$60,000	\$60,000
SD of aunt share	\$32,068	\$18,798
P(aunt within \$10k of even)	4.0% (164/4096)	40.3%
P(exactly \$60,000)	0.10% (4/4096)	—
Expected gap (aunt – uncle, absolute)	\$60,000	\$30,339
Median gap	\$60,000	\$26,000
Shape	two humps, near \$30k and \$90k	one hump, centered \$60k

Exact bin probabilities for the aunt's share (left-closed bins, \$000s):

0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80	80–90	90–100	100–110	110–120
1.8%	7.9%	14.9%	15.0%	8.4%	2.0%	1.9%	7.9%	14.9%	15.0%	8.4%	2.0%

The last bin includes the \$120,000 outcome (0.05%). The 4.3 variant's bins run 0.2, 1.2, 4.0, 9.1, 15.2, 19.7, 19.7, 15.6, 9.4, 4.3, 1.3, 0.2%.

Note. Why the expected gap is exactly \$60,000. Let R be the value of the non-cabin items the cabin loser takes. The cabin winner has $\$60,000 + (\$60,000 - R)$, the loser has R , and the gap is $\$120,000 - 2R$. By symmetry R averages $\$30,000$, so the gap averages $\$60,000$. On average the settlement is off by one entire expected share. This is the number to write on the board at the end of Part 3.

Watch out. Bin edges. Every aunt total is a multiple of \$500, and 216 of the 4096 outcomes (5.3%) land *exactly* on a bin edge (\$20k, \$30k, \$40k, \$80k, \$90k, \$100k are the common ones). Announce the convention before teams post: **a total on an edge goes in the bin to its right**, so \$60,000 goes in 60–70.

Watch out. Few marks. With teams of 4–5, a section of about 20 posts only about 10 settlements. At $n = 10$ the two humps may not both show clearly, and there is a 66% chance that *no* settlement lands within \$10k of even. The empty middle makes the point by itself, but if you have spare coins, have each team run a second aunt/uncle pair so there are 4 settlements per team. Either way, project the exact distribution after the class histogram is up.

Part 1. Predict First

(≈ 5 min)

1.1 \$60,000 is correct. Each item goes to the aunt with probability $\frac{1}{2}$, so she expects half of every appraised value. Almost everyone gets this; that's the setup, not the payoff.

1.2 Tally the circles on the board. The answer depends on how students read “how far from an even split”:

	under \$5k	\$5k–15k	\$15k–30k	over \$30k
Gap between aunt and uncle	0.3%	1.7%	7.6%	90.4%
Aunt's distance from \$60k (= gap ÷ 2)	0.9%	9.5%	40.4%	49.2%

Watch out. 1.2 is ambiguous as written, and the two readings differ by a factor of two. Part 2 defines “Gap” as the aunt–uncle difference, so use the first row as the answer key. The most common student answer is probably “under \$5,000” or “\$5,000–15,000”, which the first row puts at about 2% combined. Consider rewording the handout to “How far apart do you expect your aunt's and uncle's totals to be?”

Part 2. The Settlement

(≈ 10 min)

2.1 Varies. **Quick check:** aunt + uncle must equal \$120,000, and $\text{Gap} = |\text{aunt} - \text{uncle}| = |2 \times \text{aunt} - 120,000|$. The cabin winner is never behind; the only ties are the 4 exact \$60,000 splits.

Note. Bruno has to be flipped too, even though he adds \$0. Students will notice who “won” the dog, and 4.2 depends on them noticing.

Part 3. The Class Distribution

(≈ 12 min)

3.1 Say which “center” you mean. Class **mean** for 10 settlements: 90% of classes land between \$43k and \$77k (\$49k–71k for 24 settlements), so it will usually be near \$60k. Class **median** is unstable because the distribution is bimodal: for 24 settlements its 90% range is \$36k–84k. If the median lands in one of the humps, point that out. It’s 3.5 arriving early.

3.2 Truth: 4.0%. Expected count about 0.4 of 10 settlements and about 1 of 24. Zero is the most likely class result (66% at $n = 10$, 38% at $n = 24$).

3.3 Two humps, near \$30k and \$90k, mirror images of each other, with a near-empty valley at \$60k (bimodal, symmetric). Cause: the cabin is half the estate and equals everything else combined. The cabin flip decides which hump you are in; the other eleven items only spread you out within that hump (SD of the non-cabin share is \$11,330). Strong answers mention **both** facts: the cabin is large, and it is exactly \$60k.

3.4 Varies. The class maximum is typically about \$94k for 10 settlements (90% range \$76k–113k) and about \$102k for 24. The largest possible gap is \$120,000, one party getting every item with value (4 of 4096 settlements).

3.5 The mean is a balance point, not a typical value. Two humps at \$30k and \$90k balance at \$60k, but hardly anyone lives there. Push for the gap number: the *expected gap* is also \$60,000. The procedure is exactly even on average and, on average, off by an entire half of the estate.

Part 4. Discussion

(≈ 21 min)

4.1 Ex ante (procedural) fairness: before the flips, the rules are symmetric and each party’s expected share is \$60,000. That’s the uncle’s meaning. **Ex post** (outcome) fairness: the actual split after the coin lands. That’s the aunt’s meaning; she lost the cabin flip. Both are legitimate uses of the word, and the coin only guarantees the first.

4.2 The ring (it’s *grandmother*’s; whose grandmother?), the piano (whoever plays it), the comic books, the cabin (memories, location). An appraisal gives market value, not what each person values. A consequence worth drawing out: since the coin ignores preferences, trading after the flips could leave *both* people better off, so the coin isn’t even efficient. Bruno: a \$0 appraisal ignores attachment. Arguably his value is negative in cost terms (food, vet) and large in attachment terms.

4.3 One hump centered at \$60k. SD falls from \$32.1k to \$18.8k, P(within \$10k of even) rises from 4.0% to 40.3%, and the expected gap halves, \$60,000 → \$30,339. Fairer **ex post**; **ex ante** is unchanged (still \$60k each, still a symmetric procedure). The point: spread depends on how lumpy the items are, not only on how many there are. It’s a preview of why averages of many comparable pieces settle down.

Note. If someone asks: the extreme \$120,000 gap is still possible in the variant, just rarer (4 in 32,768 vs. 4 in 4096).

4.4 Anything defensible. Common good ones: sell everything and split the cash (requires selling, destroys sentimental value); alternating picks, draft-style (requires stated preferences and a rule for who picks first); one divides, the other chooses; bid on each item with cash payments to equalize, as in Brams & Taylor's Adjusted Winner. The graded part is the second sentence: every good answer requires the aunt and uncle to *reveal preferences, negotiate, or pay cash*, which is exactly the cooperation the coin let them skip.

4.5 From the *variance*, not bias. The procedure is unbiased, but they only get one draw from a wide, lumpy distribution, and there's no repeated trial to average it out. Fair expected value is a statement about the long run; a divorce happens once. That sentence is the takeaway for the class.

Appendix A: simulation code (instructor only)

Runs on the hub. It simulates the settlement 10,000 times; the result matches the exact enumeration above to within sampling noise (one verified run: mean \$59,982, 4.2% within \$10k; the 4.3 inventory gave \$60,299 and 40.2%). Change values to the 4.3 inventory to show the one-hump version.

```
values = make_array(60000, 14000, 11000, 8500, 7000, 5500,
                    4500, 3500, 2500, 2000, 1500, 0)

aunt_totals = make_array()
for i in np.arange(10000):
    flips = np.random.choice([0, 1], len(values))
    aunt_totals = np.append(aunt_totals, sum(flips * values))

Table().with_column("Aunt's share", aunt_totals).hist(bins=np.arange(0, 130000,
10000))
np.mean(abs(aunt_totals - 60000) <= 10000)
```

4.3 inventory:

```
values = make_array(15000, 15000, 15000, 15000, 14000, 11000, 8500,
                    7000, 5500, 4500, 3500, 2500, 2000, 1500, 0)
```

Appendix B: exact distribution (instructor only)

This code lists every possible settlement instead of simulating them. The histogram it draws is the exact distribution, and its numbers match the key: 4,096 settlements, mean 60.0, 4.0% within \$10k of even. Values are in thousands of dollars so the axis matches the handout and a bar height of 1.5 (percent per \$1k) means 15% of settlements in that \$10k bin. For the 4.3 variant, set `values = make_array(15, 15, 15, 15, 14, 11, 8.5, 7, 5.5, 4.5, 3.5, 2.5, 2, 1.5, 0)`: 32,768 settlements in one hump centered on 60, 40.3% within \$10k of even.

```

from datascience import *
import numpy as np
import matplotlib.pyplot as plots
%matplotlib inline

# Appraised values in $ thousands (matches the handout's axis)
values = make_array(60, 14, 11, 8.5, 7, 5.5,
                    4.5, 3.5, 2.5, 2, 1.5, 0)

# Start with one settlement: the aunt has nothing yet.
# For each item, every settlement so far splits in two:
# one where the aunt gets the item, one where she doesn't.
aunt_totals = make_array(0)
for i in np.arange(len(values)):
    aunt_totals = np.append(aunt_totals, aunt_totals + values.item(i))

print("Number of settlements:", len(aunt_totals))           # 4096
print("Average:", np.mean(aunt_totals))                     # 60.0
print("Within $10k of even:", np.mean(abs(aunt_totals - 60) <= 10)) # 0.04

settlements = Table().with_column("Aunt's share ($ thousands)", aunt_totals)
settlements.hist(bins=np.arange(0, 130, 10))
plots.axvline(60, color="black", linestyle="--")
plots.title("All 4096 settlements, equally likely");

```

How the algorithm works

The loop runs **12 times, once per item, not 4,096 times**. On each pass the array of settlements doubles. Every settlement built so far is copied twice: once where the aunt does not get the item (the old totals, unchanged) and once where she does (the old totals plus the item's value). `np.append(aunt_totals, aunt_totals + values.item(i))` joins those two lists. After 12 passes the array holds $2^{12} = 4096$ totals.

Trace with only the first three items (60, 14, 11):

After item	aunt_totals	Count
start	0	1
cabin (60)	0, 60	2
truck (14)	0, 60, 14, 74	4
sailboat (11)	0, 60, 14, 74, 11, 71, 25, 85	8

Each entry is one complete coin-flip outcome. For example, 25 is truck + sailboat without the cabin. Nine more items bring the count to 4,096.

Why a plain histogram is exact. Every flip is fair and independent, so all 4,096 settlements are equally likely. The fraction of the list in each bin is exactly that bin's probability. There's no sampling noise, which is the difference from Appendix A.

Limits. The list doubles with every item, so this is practical only for small inventories: 15 items give 32,768 settlements, but 40 items would give about 10^{12} . That's where simulation takes over.